Index No.: 230280L

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Github Link: https://github.com/pemitharj/EN3160_Assignment_2

Question 1

In [53]:
```
import cv2 as cv
import numpy as np
import matplotlib.pyplot as plt

im = cv.imread('images/the_berry_farms_sunflower_field.jpeg', cv.IMREAD_REDUCED_
gray = cv.cvtColor(im, cv.COLOR_BGR2GRAY).astype(np.float32) / 255.0

# Parameters
sigma_min = 4
sigma_max = 40
n_scales = 20
sigmas = np.geomspace(sigma_min, sigma_max, n_scales)

scale_space = np.zeros((gray.shape[0], gray.shape[1], n_scales), dtype=np.float3

for i, s in enumerate(sigmas):
    blurred = cv.GaussianBlur(gray, (0, 0), s)
    lap = cv.Laplacian(blurred, cv.CV_32F)
    scale_space[:, :, i] = (s**2 * lap) ** 2   # Normalized, squared so bright a

print(scale_space.shape)
print(sigmas.round(2))
```

```
(360, 360, 20)
[ 4.    4.52  5.1   5.75  6.5   7.33  8.28  9.34 10.55 11.91 13.44 15.17
 17.13 19.33 21.82 24.63 27.81 31.39 35.43 40.  ]
```

In [54]:
```
from scipy.ndimage import maximum_filter

print(scale_space.max())

# Detect extrema
threshold = 0.63 * scale_space.max()   # Tune: 0.3, 0.4, 0.5
local_max = maximum_filter(scale_space, size=(3, 3, 3)) == scale_space
mask = local_max & (scale_space > threshold)

ys, xs, si = np.where(mask)
radii = np.sqrt(2) * sigmas[si]
print(len(xs), 'blobs found')

# Suppression
responses = scale_space[ys, xs, si]
order = np.argsort(-responses)
keep = []
for i in order:
    ok = True
    for j in keep:
        d = np.hypot(xs[i] - xs[j], ys[i] - ys[j])
        if d < max(radii[i], radii[j]):
            ok = False
            break
```

```
    if ok:
        keep.append(i)

keep = np.array(keep)
bx, by, br, bs = xs[keep], ys[keep], radii[keep], sigmas[si[keep]]
print(len(bx), 'blobs after suppression')

# Remove blobs whose circle crosses the border
h, w = gray.shape
inside = (bx - br >= 0) & (bx + br < w) & (by - br >= 0) & (by + br < h)
bx, by, br, bs = bx[inside], by[inside], br[inside], bs[inside]
print(len(bx), 'blobs after removing border ones')
print((bs <= sigmas[0] + 1e-6).sum(), 'blobs at sigma_min')

# Drawing the blobs
fig, ax = plt.subplots(figsize=(6, 5))
ax.imshow(cv.cvtColor(im, cv.COLOR_BGR2RGB))
for x, y, r in zip(bx, by, br):
    ax.add_patch(plt.Circle((x, y), r, color='red', fill=False, linewidth=1.5))
ax.axis('off')
plt.show()

# Report the values
top = np.argsort(-br)[:5]
print('Sigma range used:', sigma_min, 'to', sigma_max, f'({n_scales} scales, geo
print('Largest circles (x, y, radius, sigma):')
for k in top:
    print(f'  x={bx[k]}, y={by[k]}, r={br[k]:.1f}, sigma={bs[k]:.1f}')
```

```
0.14518465
16 blobs found
16 blobs after suppression
15 blobs after removing border ones
7 blobs at sigma_min
```



```
Sigma range used: 4 to 40 (20 scales, geometric)
Largest circles (x, y, radius, sigma):
  x=282, y=338, r=10.4, sigma=7.3
  x=178, y=261, r=8.1, sigma=5.8
  x=179, y=327, r=8.1, sigma=5.8
  x=106, y=255, r=8.1, sigma=5.8
  x=63, y=243, r=6.4, sigma=4.5
```

Explanation

The sunflower image was loaded at quarter resolution (360 × 360), converted to grayscale and normalized to [0, 1]. For each σ, the image was blurred with a Gaussian and the Laplacian was applied, then multiplied by σ^2 to get the scale-normalized LoG, $\sigma^2 \nabla^2 G$. The responses were squared so that bright and dark blobs both give positive peaks. A pixel was accepted as a blob if it was a local maximum in a 3 × 3 × 3 neighbourhood (y, x, σ) and above a threshold. Overlapping detections were suppressed by keeping the strongest one, and blobs whose circles crossed the image boundary were removed. Each blob is drawn as a circle of radius $r = \sqrt{2}\,\sigma$.

Parameters

- **Scale range:** σ = 4 to 40, using 20 geometrically spaced values.
- **Threshold:** 0.63 × maximum response (≈ 0.094).
- **Neighbourhood:** 3 × 3 × 3.
- **Circle radius:** $r = \sqrt{2}\,\sigma$.

Results

The detector found 16 candidates, 16 after overlap suppression and 15 after removing boundary blobs. The largest circles are listed below (coordinates in the quarter-size image).

x (px)	y (px)	Radius (px)	σ
282	338	10.4	7.3
178	261	8.1	5.8
179	327	8.1	5.8

Discussion

At this threshold, the detections lie on sunflower heads. The largest detected scale (σ = 7.3) was well below σ_max = 40, so the upper limit was not restrictive. 7 of the 15 detections sit at σ_min = 4, so some small flowers may be clipped by the lower end of the range. A threshold of 0.2 gave 111 blobs, many on leaf texture and image borders. At 0.4, detections still appeared on tree clumps behind the field (including the two largest circles, r = 19.0 and 16.8), so I raised it to 0.63. This removes the tree detections but also misses fainter or more distant flowers, so there is a trade-off between false detections and missed flowers.

Question 2

In [55]:

```
import numpy as np
from scipy.optimize import minimize
import matplotlib.pyplot as plt

np.random.seed(0)

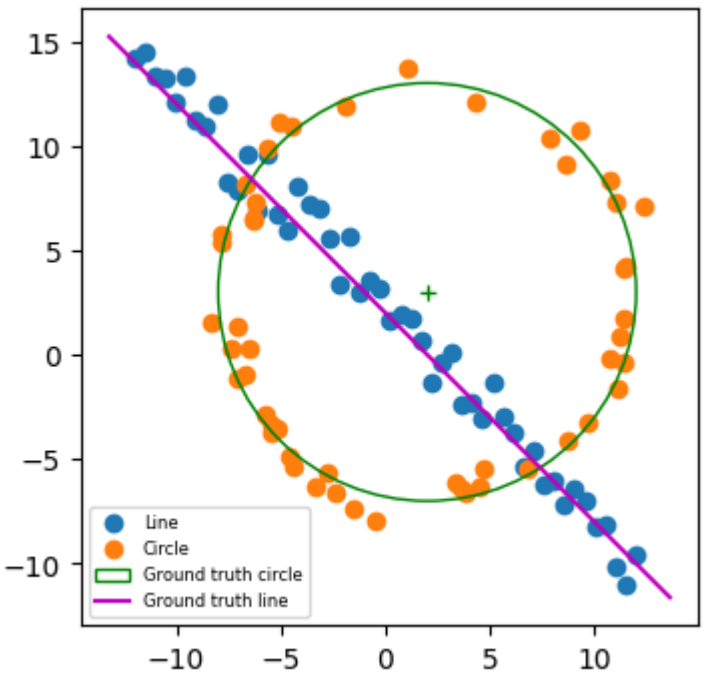
N = 100
half_n = N // 2
r = 10
x0_gt, y0_gt = 2, 3

# noisy circle
s = r / 16
t = np.random.uniform(0, 2*np.pi, half_n)
n = s * np.random.randn(half_n)
x, y = x0_gt + (r + n)*np.cos(t), y0_gt + (r + n)*np.sin(t)
X_circ = np.hstack((x.reshape(half_n, 1), y.reshape(half_n, 1)))

# noisy line
s = 1.
m, b = -1, 2
x = np.linspace(-12, 12, half_n)
y = m*x + b + s*np.random.randn(half_n)
X_line = np.hstack((x.reshape(half_n, 1), y.reshape(half_n, 1)))

X = np.vstack((X_circ, X_line))

fig, ax = plt.subplots(figsize=(4, 4))
ax.scatter(X_line[:, 0], X_line[:, 1], label='Line')
ax.scatter(X_circ[:, 0], X_circ[:, 1], label='Circle')
ax.add_patch(plt.Circle((x0_gt, y0_gt), r, color='g', fill=False, label='Ground
ax.plot(x0_gt, y0_gt, '+', color='g')
x_ = np.array(ax.get_xlim())
ax.plot(x_, m*x_ + b, color='m', label='Ground truth line')
ax.set_aspect('equal')
ax.legend(fontsize=6, loc='lower left')
plt.show()
```



(a) RANSAC line

In [56]:
```
def line_from_two_points(p1, p2):
    dx, dy = p2 - p1
    nrm = np.hypot(dx, dy)
    a, b = -dy / nrm, dx / nrm   # Unit normal
    d = a * p1[0] + b * p1[1]
    return a, b, d

def fit_line_tls(P):
    c = P.mean(axis=0)
    _, _, Vt = np.linalg.svd(P - c)
    a, b = Vt[-1]   # Unit normal by construction
    d = a * c[0] + b * c[1]
    return a, b, d

def line_error(P, a, b, d):
    return np.abs(a * P[:, 0] + b * P[:, 1] - d)   # Normal distance

rng = np.random.default_rng(0)

line_thresh = 1.0
line_min_inliers = 40
n_iter = 200

best_count = 0
best_line_inliers = None
best_line_sample = None
best_line_sample_model = None

for _ in range(n_iter):
    idx = rng.choice(len(X), 2, replace=False)
    a, b_, d = line_from_two_points(X[idx[0]], X[idx[1]])
    inl = line_error(X, a, b_, d) < line_thresh
    if inl.sum() > best_count:
        best_count = inl.sum()
        best_line_inliers = inl
        best_line_sample = X[idx]
        best_line_sample_model = (a, b_, d)

assert best_count >= line_min_inliers
a_l, b_l, d_l = fit_line_tls(X[best_line_inliers])

print('best consensus:', best_count, '(min required:', line_min_inliers, ')')
print('RANSAC line: a=%.3f b=%.3f d=%.3f, norm=%.3f' % (a_l, b_l, d_l, np.hypot(
print('ground truth: a=%.3f b=%.3f d=%.3f' % (1/np.sqrt(2), 1/np.sqrt(2), 2/np.s
```

```
best consensus: 44 (min required: 40 )
RANSAC line: a=0.725 b=0.689 d=1.782, norm=1.000
ground truth: a=0.707 b=0.707 d=1.414
```

Explanation

The best consensus contained 44 inliers, exceeding the minimum of 40. The fitted line parameters were ((a,b,d)=(0.725, 0.689, 1.782)), close to the ground truth ((0.707, 0.707, 1.414)). The small difference is due to noise and a few circle points near the intersections being included as inliers.

(b) RANSAC circle on the remnant

In [57]:
```
def circle_from_three_points(p1, p2, p3):
    A = np.array([[p1[0], p1[1], 1],
                  [p2[0], p2[1], 1],
                  [p3[0], p3[1], 1]])
    rhs = -np.array([p1 @ p1, p2 @ p2, p3 @ p3])
    try:
        D, E, F = np.linalg.solve(A, rhs)
    except np.linalg.LinAlgError:
        return None
    cx, cy = -D / 2, -E / 2
    return cx, cy, np.sqrt(cx**2 + cy**2 - F)

def circle_error(P, cx, cy, rad):
    return np.abs(np.hypot(P[:, 0] - cx, P[:, 1] - cy) - rad)   # Radial error

def fit_circle_opt(P, init):
    cost = lambda p: np.sum(circle_error(P, p[0], p[1], p[2])**2)
    return minimize(cost, init).x

X_rem = X[~best_line_inliers]

circ_thresh = 1.0
circ_min_inliers = 35
n_iter_c = 500

best_count_c = 0
best_circ_inliers = None
best_circ_sample = None
best_circ_sample_model = None

for _ in range(n_iter_c):
    idx = rng.choice(len(X_rem), 3, replace=False)
    model = circle_from_three_points(*X_rem[idx])
    if model is None:
        continue
    inl = circle_error(X_rem, *model) < circ_thresh
    if inl.sum() > best_count_c:
        best_count_c = inl.sum()
        best_circ_inliers = inl
        best_circ_sample = X_rem[idx]
        best_circ_sample_model = model

assert best_count_c >= circ_min_inliers
cx, cy, rc = fit_circle_opt(X_rem[best_circ_inliers], best_circ_sample_model)

print('remaining points:', len(X_rem))
print('best consensus:', best_count_c, '(min required:', circ_min_inliers, ')')
print('RANSAC circle: centre=(%.2f, %.2f), r=%.2f' % (cx, cy, rc))
print('ground truth : centre=(2, 3), r=10')
```

```
remaining points: 56
best consensus: 45 (min required: 35 )
RANSAC circle: centre=(1.82, 3.03), r=9.82
ground truth : centre=(2, 3), r=10
```

Explanation

The best consensus contained 45 inliers, exceeding the minimum required of 35. The fitted circle had centre = (1.82, 3.03) and radius = 9.82, close to the ground truth centre = (2, 3) and radius = 10. The small differences are due to noise and possible inclusion of a few outliers.

(c) Combined Plots

In [58]:
```
fig, ax = plt.subplots(figsize=(5, 5))

ax.scatter(X[:, 0], X[:, 1], c='lightgray', label='All points')
ax.scatter(*X[best_line_inliers].T, c='tab:blue', label='Line inliers')
ax.scatter(*X_rem[best_circ_inliers].T, c='tab:orange', label='Circle inliers')

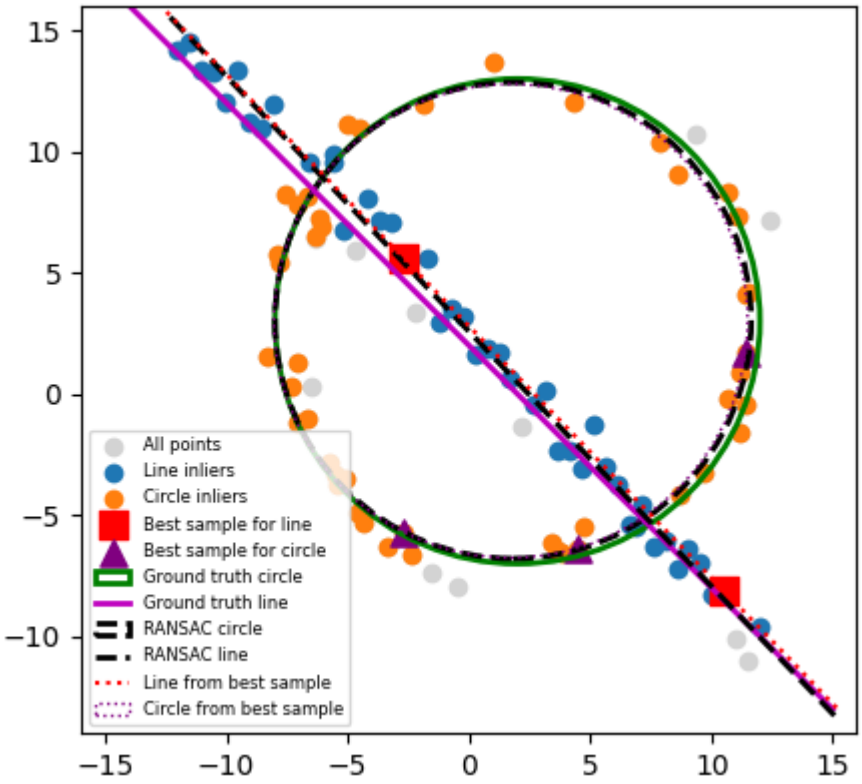
# Best samples
ax.scatter(*best_line_sample.T, c='red', marker='s', s=90, label='Best sample fo
ax.scatter(*best_circ_sample.T, c='purple', marker='^', s=90, label='Best sample

# Ground truth
ax.add_patch(plt.Circle((x0_gt, y0_gt), r, color='g', fill=False, linewidth=2, l
xs_ = np.array([-15, 15])
ax.plot(xs_, m*xs_ + b, color='m', linewidth=2, label='Ground truth line')

# RANSAC results
ax.add_patch(plt.Circle((cx, cy), rc, color='k', fill=False, linestyle='--', lin
tt = np.linspace(-20, 20, 2)
p0 = d_l * np.array([a_l, b_l]); dirn = np.array([-b_l, a_l])
L = p0 + tt[:, None] * dirn
ax.plot(L[:, 0], L[:, 1], 'k--', linewidth=2, label='RANSAC line')

# Models from the best samples only
a_s, b_s, d_s = best_line_sample_model
p0s = d_s * np.array([a_s, b_s]); dirs = np.array([-b_s, a_s])
Ls = p0s + tt[:, None] * dirs
ax.plot(Ls[:, 0], Ls[:, 1], color='red', linestyle=':', label='Line from best sa
ax.add_patch(plt.Circle(best_circ_sample_model[:2], best_circ_sample_model[2],
                        color='purple', fill=False, linestyle=':', label='Circle

ax.set_xlim(-16, 16); ax.set_ylim(-14, 16)
ax.set_aspect('equal')
ax.legend(fontsize=6, loc='lower left')
plt.show()
```



(d) What will happen if we fit the circle first?

In [59]:

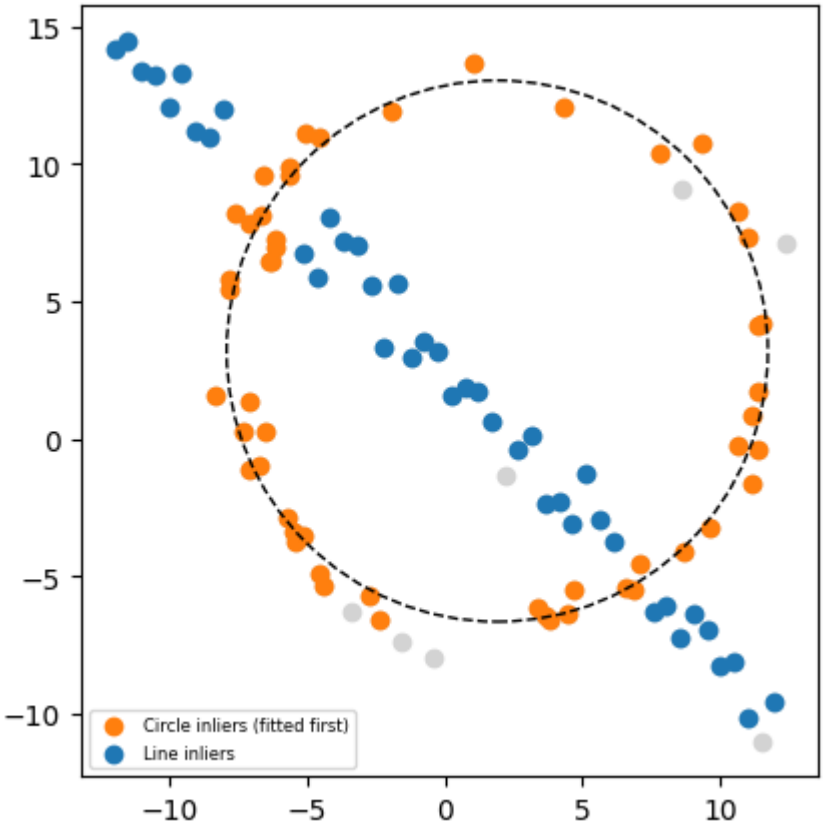
```
rng2 = np.random.default_rng(1)
best_c, best_inl_c, best_model_c = 0, None, None
for _ in range(500):
    idx = rng2.choice(len(X), 3, replace=False)
    model = circle_from_three_points(*X[idx])
    if model is None:
        continue
    inl = circle_error(X, *model) < circ_thresh
    if inl.sum() > best_c:
        best_c, best_inl_c, best_model_c = inl.sum(), inl, model

cx2, cy2, rc2 = fit_circle_opt(X[best_inl_c], best_model_c)
print('circle-first: consensus', best_c, '| centre=(%.2f, %.2f), r=%.2f' % (cx2,

X_rem2 = X[~best_inl_c]
best_l, best_inl_l = 0, None
for _ in range(200):
    idx = rng2.choice(len(X_rem2), 2, replace=False)
    a, b_, d = line_from_two_points(X_rem2[idx[0]], X_rem2[idx[1]])
    inl = line_error(X_rem2, a, b_, d) < line_thresh
    if inl.sum() > best_l:
        best_l, best_inl_l = inl.sum(), inl
a2, b2, d2 = fit_line_tls(X_rem2[best_inl_l])
print('line after circle: consensus', best_l, '| a=%.3f b=%.3f d=%.3f' % (a2, b2

fig, ax = plt.subplots(figsize=(5, 5))
ax.scatter(X[:, 0], X[:, 1], c='lightgray')
ax.scatter(*X[best_inl_c].T, c='tab:orange', label='Circle inliers (fitted first
ax.scatter(*X_rem2[best_inl_l].T, c='tab:blue', label='Line inliers')
ax.add_patch(plt.Circle((cx2, cy2), rc2, color='k', fill=False, linestyle='--'))
ax.set_aspect('equal'); ax.legend(fontsize=6, loc='lower left')
plt.show()
```

```
circle-first: consensus 52 | centre=(1.92, 3.20), r=9.84
line after circle: consensus 41 | a=-0.721 b=-0.693 d=-1.704
```



Explanation

- RANSAC detected 52 circle inliers, with centre = (1.92, 3.20) and radius = 9.84, which are close to the ground truth (2, 3) and 10.
- So, some line points near the two crossing points were wrongly counted as circle inliers, slightly shifting the estimated centre and radius.
- The line fitted on the remaining points had 41 inliers, compared to 44 when the line was fitted first, because some line points were absorbed by the circle.
- However, the effect is small in this experiment, but it would be greater with more noise, a larger threshold, or a tangent line.
- So, fitting the line first is safer because it requires only 2 points per sample, and the first model can claim shared points in sequential RANSAC.

Question 3

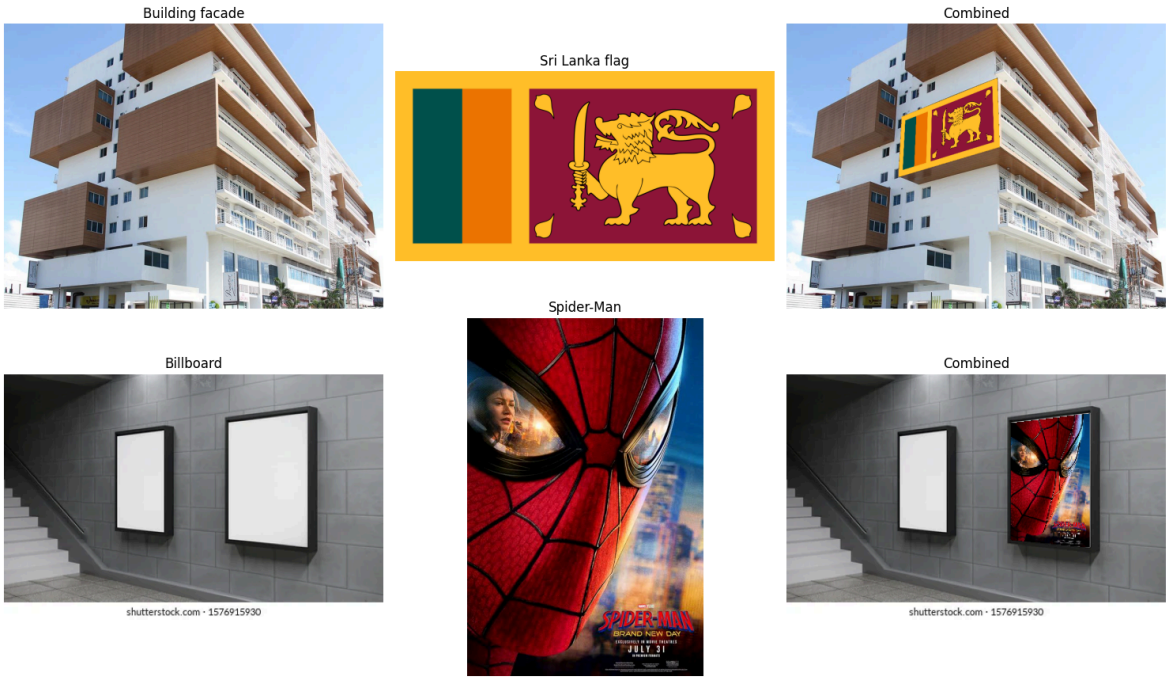
In [60]:
```python
def click4(path):
    # Click 4 corners: top-left, top-right, bottom-right, bottom-left
    img, pts = cv.imread(path), []
    def cb(e, x, y, f, p):
        if e == cv.EVENT_LBUTTONDOWN:
            pts.append([x, y]); cv.circle(img, (x, y), 5, (0, 0, 255), -1)
    cv.namedWindow('click'); cv.setMouseCallback('click', cb)
    while len(pts) < 4 and cv.waitKey(20) != 27:
        cv.imshow('click', img)
    cv.destroyAllWindows()
    return pts
```

```
def overlay(bg_path, fg_path, pts):
    bg, fg = cv.imread(bg_path), cv.imread(fg_path)
    h, w = fg.shape[:2]
    H, _ = cv.findHomography(np.float32([[0,0],[w,0],[w,h],[0,h]]), np.float32(p
    size = (bg.shape[1], bg.shape[0])
    warped = cv.warpPerspective(fg, H, size)
    mask = cv.warpPerspective(np.full((h, w), 255, np.uint8), H, size) > 0
    out = bg.copy(); out[mask] = warped[mask]
    return out
```

In [61]:
```
pairs = [('images/building facade.jpg', 'images/sri_lanka_flag.webp', 'Building
         ('images/billboard.webp', 'images/spiderman.jpg', 'Billboard', 'Spider-

fig, axs = plt.subplots(2, 3, figsize=(15, 9))
for row, (bg, fg, t1, t2) in zip(axs, pairs):
    pts = click4(bg)
    print(bg, pts)
    imgs = [cv.imread(bg), cv.imread(fg), overlay(bg, fg, pts)]
    for ax, im, t in zip(row, imgs, [t1, t2, 'Combined']):
        ax.imshow(cv.cvtColor(im, cv.COLOR_BGR2RGB)); ax.set_title(t); ax.axis('
plt.tight_layout(); plt.show()
```

```
images/building facade.jpg [[277, 223], [507, 130], [513, 296], [269, 368]]
images/billboard.webp [[256, 56], [346, 45], [346, 197], [256, 189]]
```



Explanation

- Pair 1 uses a building facade with the Sri Lankan flag, so the overlay demonstrates how the image fits a flat surface viewed at an angle.
- Pair 2 uses a billboard with a Spider-Man image, as billboards provide clear rectangular surfaces for image replacement.
- In both results, the overlaid images follow the perspective of the surfaces. However, small corner-selection errors or differences in lighting can make the overlays look less realistic.

Question 4

(a) SIFT Features

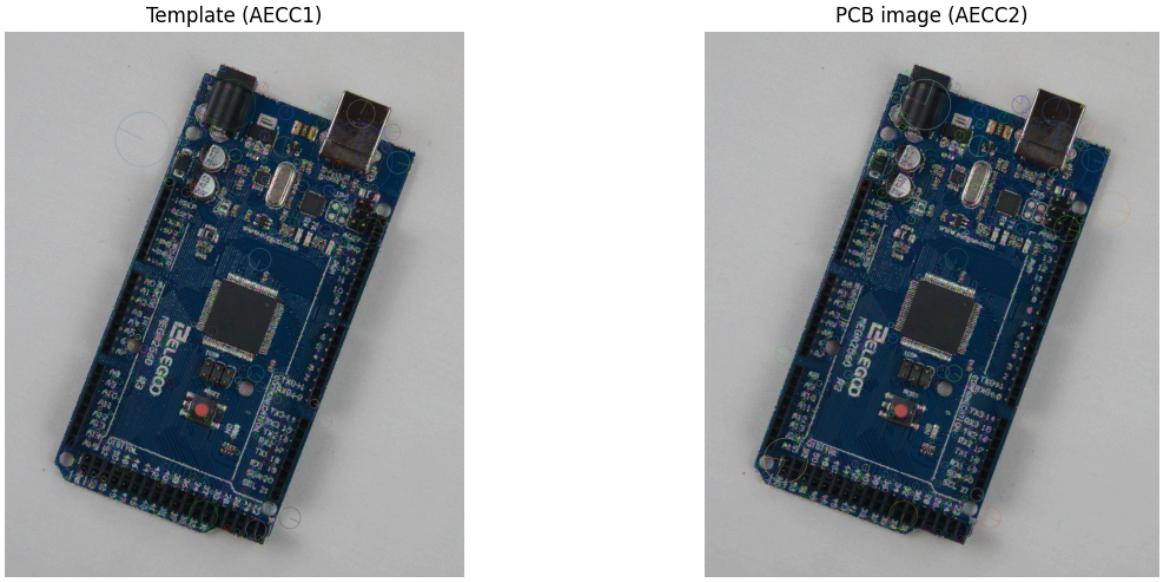
In [62]:
```
import cv2 as cv, numpy as np, matplotlib.pyplot as plt

template = cv.imread('images/pcbs/AECC1.jpg')
pcb = cv.imread('images/pcbs/AECC2.jpg')   # Example PCB image for parts (a) to

sift = cv.SIFT_create()
kp_t, des_t = sift.detectAndCompute(cv.cvtColor(template, cv.COLOR_BGR2GRAY), No
kp_p, des_p = sift.detectAndCompute(cv.cvtColor(pcb, cv.COLOR_BGR2GRAY), None)
print('template keypoints:', len(kp_t), '| PCB keypoints:', len(kp_p))

fig, axs = plt.subplots(1, 2, figsize=(14, 6))
for ax, im, kp, t in zip(axs, [template, pcb], [kp_t, kp_p], ['Template (AECC1)'
    ax.imshow(cv.cvtColor(cv.drawKeypoints(im, kp, None, flags=cv.DRAW_MATCHES_F
    ax.set_title(t); ax.axis('off')
plt.show()
```

```
template keypoints: 14272 | PCB keypoints: 12315
```



SIFT (`cv2.SIFT_create()`) was used to detect keypoints and compute descriptors in the template (AECC1) and in each PCB image. SIFT features are robust to scale and rotation, so they can be matched even when the board is photographed from a different position.

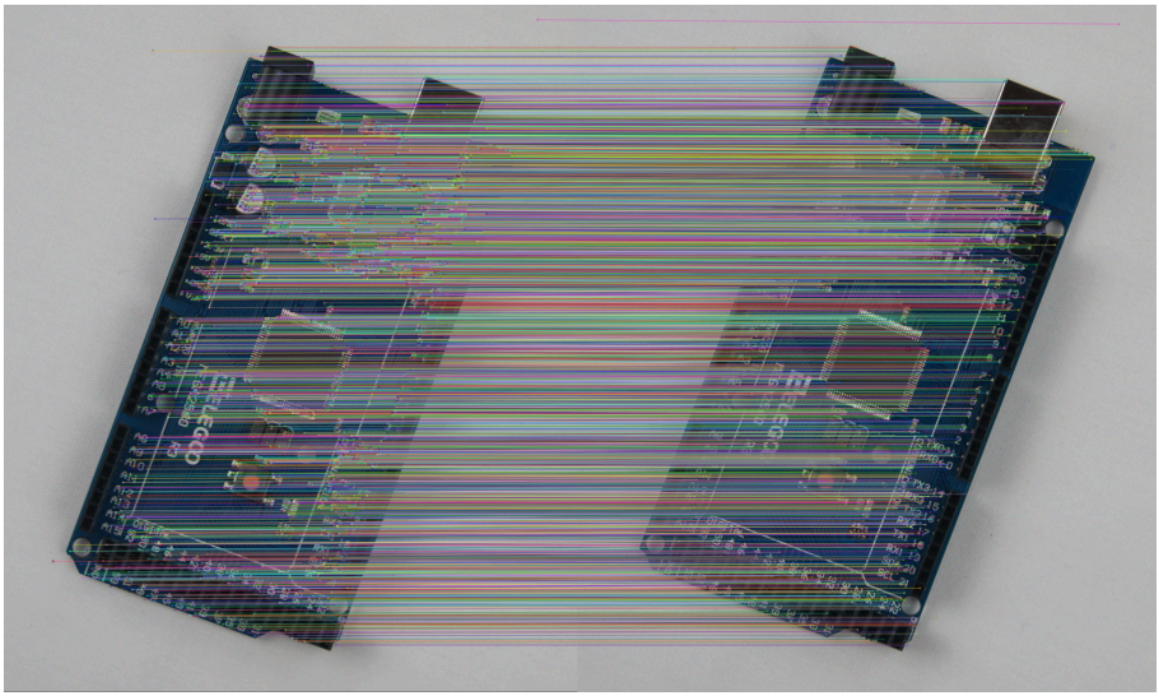
(b) Match the features

In [63]:
```
bf = cv.BFMatcher()
knn = bf.knnMatch(des_p, des_t, k=2)
good = [m for m, n in knn if m.distance < 0.75 * n.distance]   # Lowe's ratio te
print('good matches:', len(good))

src = np.float32([kp_p[m.queryIdx].pt for m in good]).reshape(-1, 1, 2)   # PCB
dst = np.float32([kp_t[m.trainIdx].pt for m in good]).reshape(-1, 1, 2)   # Temp
H, mask = cv.findHomography(src, dst, cv.RANSAC, 5.0)
print('RANSAC inliers:', int(mask.sum()), '| H =\n', H)

vis = cv.drawMatches(pcb, kp_p, template, kp_t, good, None,
                     matchesMask=mask.ravel().tolist(), flags=cv.DrawMatchesFlag
plt.figure(figsize=(16, 7)); plt.imshow(cv.cvtColor(vis, cv.COLOR_BGR2RGB)); plt
```

```
good matches: 4322
RANSAC inliers: 3847 | H =
 [[ 1.00057628e+00 -1.50450714e-02  3.21100381e+01]
 [ 1.65924885e-02  9.99995224e-01 -1.81596023e+01]
 [ 3.48317715e-07  2.46480108e-07  1.00000000e+00]]
```

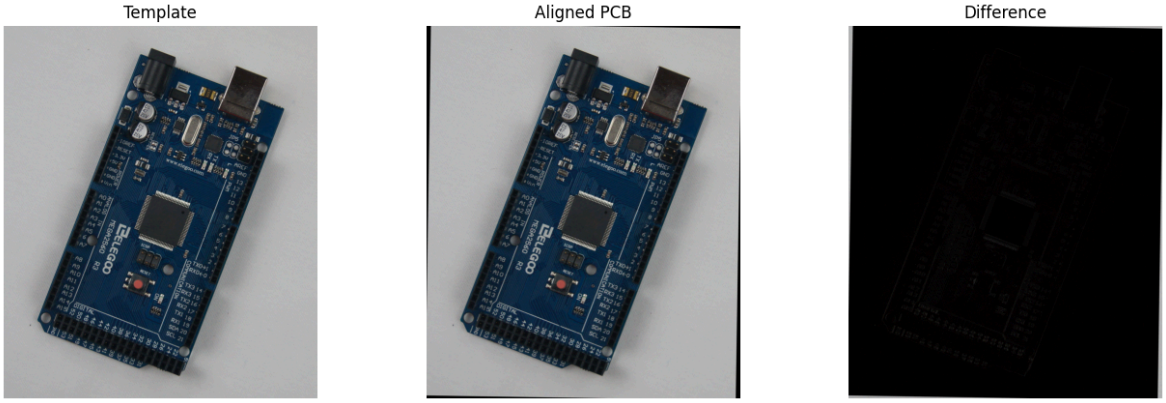


Descriptors were matched with `cv2.BFMatcher()` using k-nearest neighbours, and only matches passing Lowe's ratio test (distance < 0.75 × second-best) were kept. The homography from each PCB image to the template was then estimated with RANSAC (reprojection threshold 5 px), which rejects wrong matches. The matches figure shows only the RANSAC inliers.

(c) Warp and difference image

In [64]:
```
h, w = template.shape[:2]
aligned = cv.warpPerspective(pcb, H, (w, h))
diff = cv.absdiff(aligned, template)

fig, axs = plt.subplots(1, 3, figsize=(16, 5))
for ax, im, t in zip(axs, [template, aligned, diff], ['Template', 'Aligned PCB',
    ax.imshow(cv.cvtColor(im, cv.COLOR_BGR2RGB)); ax.set_title(t); ax.axis('off'
plt.show()
```



The PCB image was warped to the template's frame with `cv2.warpPerspective` using the estimated homography. The absolute difference between the aligned image and the template (`cv2.absdiff`) was then computed. Where the alignment is good, the difference image is mostly dark.

(d) All PCB images

In [65]:
```
bf = cv.BFMatcher()
h, w = template.shape[:2]

names = ['AAAA1', 'ADAD1', 'AECC2', 'AECC3', 'AECC4']
fig, axs = plt.subplots(len(names), 3, figsize=(15, 4.5 * len(names)))

for row, n in zip(axs, names):
    img = cv.imread(f'images/pcbs/{n}.jpg')
    kp, des = sift.detectAndCompute(cv.cvtColor(img, cv.COLOR_BGR2GRAY), None)

    good = [m for m, k in bf.knnMatch(des, des_t, k=2) if m.distance < 0.75 * k.
    src = np.float32([kp[m.queryIdx].pt for m in good]).reshape(-1, 1, 2)
    dst = np.float32([kp_t[m.trainIdx].pt for m in good]).reshape(-1, 1, 2)
    H, mask = cv.findHomography(src, dst, cv.RANSAC, 5.0)

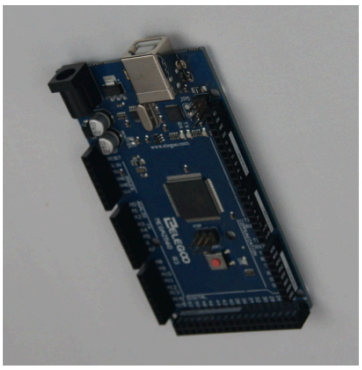
    aligned = cv.warpPerspective(img, H, (w, h))
    diff = cv.absdiff(aligned, template)
    print(n, '| good matches:', len(good), '| RANSAC inliers:', int(mask.sum()))

    for ax, im, t in zip(row, [img, aligned, diff], [n, 'Aligned', 'Difference']
        ax.imshow(cv.cvtColor(im, cv.COLOR_BGR2RGB))
        ax.set_title(t); ax.axis('off')

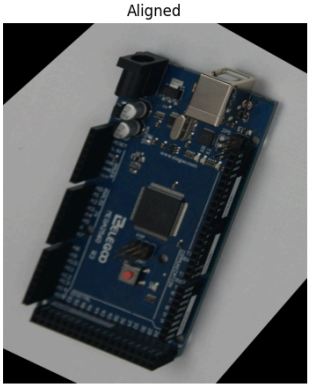
plt.tight_layout(); plt.show()
```

```
AAAA1 | good matches: 51 | RANSAC inliers: 18
ADAD1 | good matches: 369 | RANSAC inliers: 194
AECC2 | good matches: 4322 | RANSAC inliers: 3847
AECC3 | good matches: 5629 | RANSAC inliers: 5006
AECC4 | good matches: 3830 | RANSAC inliers: 3402
```

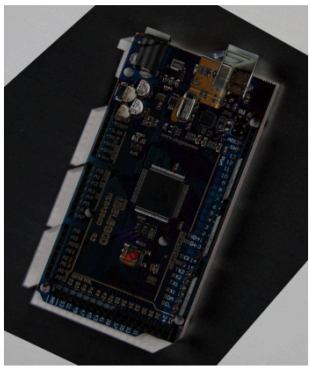
AAAA1



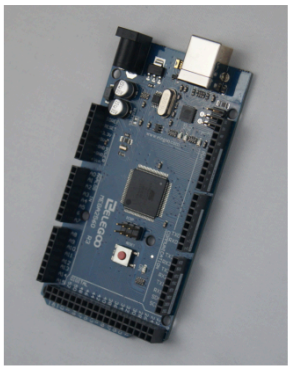
Aligned



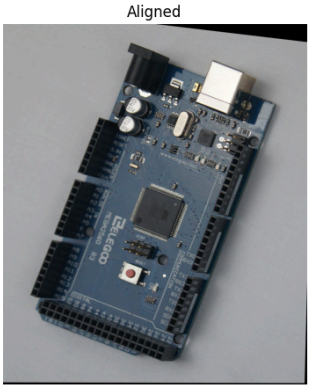
Difference



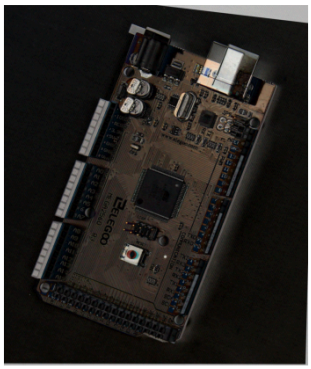
ADAD1



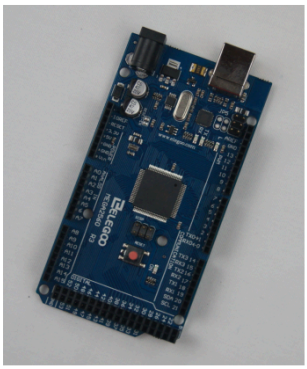
Aligned



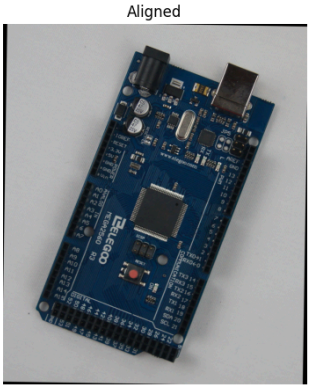
Difference



AECC2



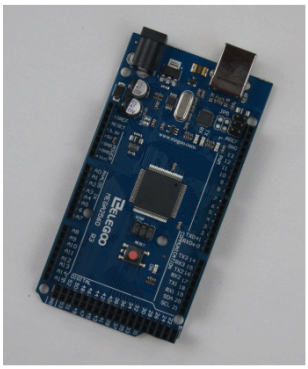
Aligned



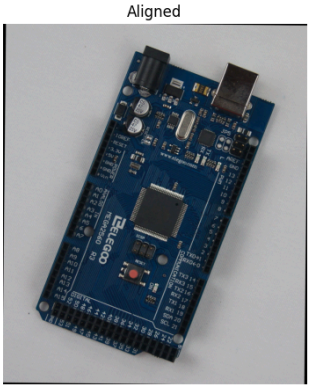
Difference



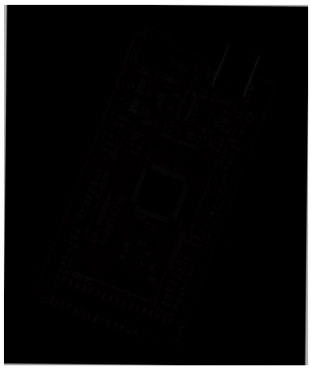
AECC3



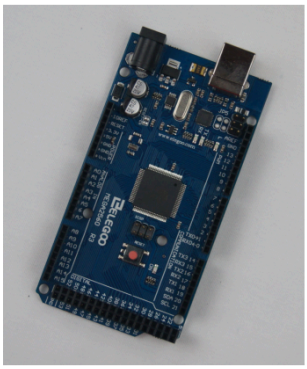
Aligned



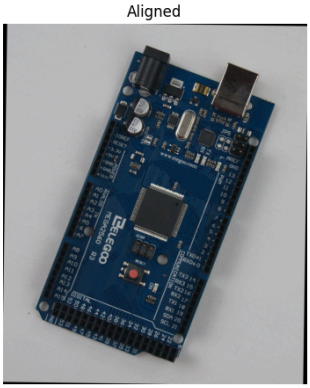
Difference



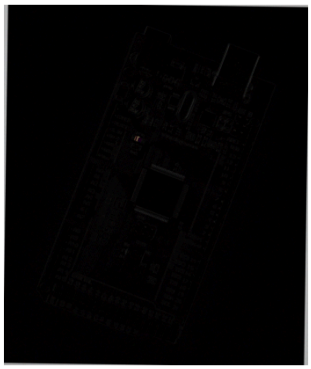
AECC4



Aligned



Difference



The same pipeline was applied to all five images. The alignment was strong for AECC2, AECC3 and AECC4 (about 89% inliers, 3402 to 5006 inliers). ADAD1 gave 194 inliers (53%), and its board looks lighter in colour than the template, so the SIFT descriptors match less well and the difference image shows a brightness offset even where the alignment is good. AAAA1 was the weakest (18 inliers out of 51 matches, 35%) because it was photographed from a very different viewpoint angle. SIFT is only partly invariant to large perspective changes, so fewer reliable matches survive the ratio test, and the homography is less accurate.

(e) Why do the USB Type B port, capacitors and reset button show high difference?

Even though the board is aligned well, these components still show high differences for several reasons:

1. **Parallax:** The homography aligns the flat PCB surface, but these components protrude above it. So, they shift relative to the board when the camera position changes.
2. **Reflective surfaces:** The USB port and capacitor tops reflect light differently depending on the viewing angle, so their pixel values change even when they are approximately aligned.
3. **Lighting and shadows:** Differences in illumination, exposure and shadows around these raised components also increase the pixel differences.
4. **Alignment errors:** These components have sharp, high-contrast edges, so even a small misalignment of a few pixels produces a large difference.

So, the flat PCB regions show lower differences because the homography aligns them more accurately.